

Unplugged Coin Addition Activity Description

Activity Description: This unplugged activity simulates the addition of all elements in an array. The purpose of this activity is for students to recognize a situation in which there is a race condition because of a shared resource (the accumulator storage location in this example).

Prerequisites: It is helpful but not necessary if students have done the Unplugged Coin Search Activity. It is also helpful but not necessary that students have been exposed to array syntax to sum the contents of an array using an accumulation loop. It may additionally be motivating for students to have tried to naively parallelize this activity as shown below using OpenMP and discovered that it doesn't work.

Materials Needed: Containers with the same number of coins of different denominations (pennies, nickels, dimes, quarters),

Description of the activity: Students should be in groups of 4-5. Each group should be given a cup with coins and empty containers to use to distribute coins to team members. Students should be asked how to distribute the work of figuring out how much money is in the group in total. Students will recognize that dividing the coins among group members and having each student count their container should decrease the time needed to figure out the cumulative sum. However, students should be reminded that the loop to add the contents of an array, looks like:

```
total = 0
for each value in the array
    total = total + value
```

The important line of code to point out is that before the current element of the array can be added to `total`, the process needs to have access to the current value of `total`. To simulate this part of the program, have one student (or the instructor if the class is small) act as the keeper memory location for `total`, in charge of giving the `total` to each process (student) and recording the new value as each student accesses it. Each process needs to request the current contents of `total`, add the current value to it and then store it back in `total`.

Students should hopefully observe that `total` is not being accumulated correctly as each student counter competes to get the current contents of `total` and then store their updated value.

Instructor Notes:

- Online-synchronous student who could participate if warned to have the materials.
- Online-asynchronous students may benefit from a video recorded of the activity

- Foreign student who was not familiar with US currency. (Maybe notecards with numbers on them would be better)